

Surface Graft Copolymerization of Viologens on Polymeric Substrates

S. W. Ng, K. G. Neoh,* Y. T. Wong, J. T. Sampanthar, and E. T. Kang

Department of Chemical and Environmental Engineering, National University of Singapore, Kent Ridge, Singapore 119260

K. L. Tan

Department of Physics, National University of Singapore, Kent Ridge, Singapore 119260

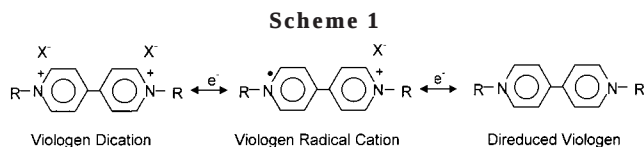
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Preparation of pendent-type viologens on polymeric substrates was carried out by a surface graft copolymerization technique. This method involved a two-step process whereby vinyl benzyl chloride (VBC) was first graft copolymerized on the polymeric substrate followed by the attachment of viologen moieties to the graft-copolymerized benzyl chloride groups. The graft copolymerization was carried out by placing the plasma pretreated polymeric substrate in contact with VBC under UV irradiation. The effects of plasma pretreatment time, UV graft copolymerization time, solvent, and monomer concentration on the VBC graft copolymer concentration on the films were studied. Two different methods for the subsequent formation of viologens were also attempted. The graft-modified films were characterized using atomic force microscopy, X-ray photoelectron spectroscopy, and UV–visible absorption spectroscopy. The response of the viologen-grafted films to photoirradiation and the bleaching of the irradiated films were investigated and found to be dependent on the viologen graft concentration, the counteranion associated with the viologen, and the light intensity.

Introduction

Chromogenic technology is an actively pursued area of research for the development of materials and devices in the modulation of optical or thermal energy. Commonly used chromogenic materials include the transition metal oxides such as WO_3 and organic compounds such as viologens.^{1–4} The color change in the former occurs as a result of dual injection (or ejection) of electrons and ions and is frequently driven by the application of an electrical potential. The devices based on these materials may be constructed as multilayer devices, for example, incorporating two outer transparent electrical conductors, the electrochromic material, and an electrolyte or ion conductor.

In the case of the viologens or 1,1'-disubstituted-4,4'-bipyridiniums, the change in coloration is achieved by the oxidation–reduction reactions shown in Scheme 1. A one-electron reduction of the lightly colored viologen dication results in the intensely colored radical cation.^{5–7} This color change may be initiated by near-UV irradiation, electrical current, or a chemical reductant. Viologens with short alkyl chain substituents are soluble in water, and to circumvent this problem, studies have been carried out with viologens incorporated into anionic polyelectrolyte films or other matrix polymers,^{5,8–9} in zeolite matrixes,^{10,11}



and in iminodiacetic acid-type chelate resin beads.¹² In other cases, polymers with viologen moieties have been synthesized.^{13–17} In these techniques, the viologen moieties are part of or evenly distributed throughout the host matrix.

In the present work, we report a novel approach of preparing pendent-type viologen polymers on polymeric substrates by surface graft copolymerization. The viologen polymeric layer is thus confined to the surface of the substrate. The surface modification of polymeric substrates via graft copolymerization to impart new and specific functionalities has been previously carried out by our group as well as others.^{18–22} However, there is little

* To whom correspondence should be addressed. Tel: +65 8742186. Fax: +65 7791936. E-mail: chenkg@nus.edu.sg.

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Scheme 2**LDPE films**Argon plasma
pretreatment**Graft copolymerization of VBC on plasma pretreated films**

- a) Solution grafting technique
- b) Quartz plate grafting technique

Formation of viologen moieties on VBC-graft copolymerized films

- a) 2-step method
- b) 1-step method

Viologen-grafted LDPE films

work thus far on the grafting of viologen on the surface of polymers.^{23,24} Our studies presented in this paper are conducted primarily on low-density polyethylene (LDPE) films. The viologen graft copolymerized films were characterized by atomic force microscopy (AFM), X-ray photoelectron spectroscopy (XPS), and UV-visible absorption spectroscopy. The effects of the reaction conditions such as the temperature, solvent, and monomer concentration used in the graft copolymerization process and the response of these graft-modified films to photoirradiation were evaluated.

Experimental Section

Sample Preparation. The grafting of viologens on LDPE films was achieved through the initial graft copolymerization of vinyl benzyl chloride (VBC) on the LDPE substrates, followed by the chemical reaction of the grafted groups with 4,4'-bipyridine and alkyl halides to form viologen moieties. Different methods for the graft copolymerization of VBC on LDPE as well as for the subsequent reactions to form viologens were studied. The reaction steps are summarized in Scheme 2 and will be further elaborated.

LDPE films (0.125 mm in thickness) obtained from Goodfellow, Inc., were washed in acetone for 1 h using an ultrasonic bath to remove surface impurities. The films were dried under reduced pressure and cut into strips of 2 cm × 5 cm. The washed films were argon plasma treated for a predetermined period of time on both sides, at an argon pressure of approximately 0.6 Torr, using an Anatech SP100 plasma system. These plasma-treated films were further exposed to air for approximately 5 min to facilitate the formation of surface oxide and peroxide groups²⁵ before graft copolymerization of VBC was carried out. Two techniques for the graft copolymerization of VBC on LDPE were studied. The solution grafting technique involved placing the argon plasma pretreated films in degassed solutions of VBC. Ethanol, 2-propanol, xylene, tetrahydrofuran (THF), and dimethylformamide (DMF) were used as solvents. Degassing of the solutions was achieved by bubbling argon vigorously into the solutions in test tubes for 40 min before stoppering and sealing the tubes. The tubes of VBC solutions containing the LDPE films were then exposed to UV irradiation in a Riko rotary photochemical reactor (RH400-10W) for 2 h at 28 °C. The concentration of VBC was varied from 1 to 10 vol %. The graft copolymerized films were finally subjected to prolonged washing with DMF to remove residual homopolymers and reactants. On the other hand,

the quartz plate grafting technique was carried out by placing the monomer (97% purity) on both surfaces of the argon plasma pretreated LDPE film, sandwiching the film between two pieces of quartz plates, and exposing the assembly to near-UV irradiation for different periods of time. The graft copolymerized films were extracted from the quartz plates after prolonged immersion in DMF and then again subjected to thorough washing with DMF.

Two methods were also investigated for the formation of viologen moieties on the VBC-graft copolymerized films. The first method (method 1) involved a two-step reaction of the films with 4,4'-bipyridine (1 wt % in DMF), followed by benzyl chloride according to the Menshutkin reaction.⁶ The reaction of the films in 4,4'-bipyridine was allowed to proceed in a water bath at a temperature of between 40 and 90 °C for various periods of time. The reacted films were washed in DMF to remove unreacted reactants before being placed in 5 vol % of benzyl chloride in DMF. This reaction was also allowed to proceed at different temperatures and for different periods of time. The reacted films were finally washed in DMF and rinsed with deionized water. In the second method (method 2), a one-step reaction of the VBC-graft copolymerized films in a solution mixture containing dichloro-*p*-xylene and 4,4'-bipyridine was carried out in a manner similar to the preparation of ionene-type viologen polymers.^{6,26} An equimolar mixture of dichloro-*p*-xylene and 4,4'-bipyridine (0.024 M of each reactant) was prepared using DMF as solvent. The reaction of the films in the solution mixture was carried out at 60 °C for 20 h. The reacted films were washed with DMF followed by deionized water to remove residual reactants and homopolymers. The viologen-grafted films were finally dried under reduced pressure and stored in the dark until further studies. To study the effect of anions on the response of the viologen-grafted films, ion exchange was carried out by immersing the films in a 25 wt % solution of poly(vinylsulfonic acid, sodium salt).

Testing and Characterization. The irradiation of the viologen films was performed at a temperature of between 25 and 28 °C, using a 1-kW Hg lamp in the Riko rotary reactor. The samples were placed in a Pyrex tube and positioned 5 cm from the light source. The Pyrex tube cuts off wavelengths below 290 nm completely, with approximately 50% transmission at 305 nm. For investigations on the effect of light intensity on the samples, optical density filters (from Ealing Corp.) were used to reduce the transmission by 50% and 80%. In the experiments requiring the absence of air, the films were placed in an evacuated quartz cell.

The UV-visible absorption spectra of the films were obtained using a Shimadzu UV-3101 PC scanning spectrometer, with pristine LDPE films as reference. XPS analysis of the films was made on a VG ESCALAB MkII spectrometer with a Mg K α X-ray source (1253.6-eV photons). The X-ray source was run at a power of 120 W (12 kV and 10 mA). The pressure in the analysis chamber was maintained at 10⁻⁸ mbar or lower during measurements. The polymer films were mounted on standard VG sample studs by means of double-sided adhesive tape. To compensate for surface charging effect, all core-level spectra were referenced to the C 1s hydrocarbon peak at 284.6 eV. In spectral deconvolution, the full width at half-maximum (fwhm) was maintained constant for all components in a particular spectrum. The peak area ratios for the various elements were corrected using experimentally determined instrumental sensitivity factors and are accurate to $\pm 10\%$. AFM studies of the films were carried out using a Nanoscope III scanning atomic force microscope. All images were scanned in air under constant force mode, with a scan rate of 1 Hz and a scan size of 1 μm .

Results and Discussion

Graft Copolymerization of VBC. The success of surface graft copolymerization of VBC on the LDPE substrates using both solution and quartz plate grafting techniques is ascertained by the XPS Cl 2p spectra of the films after copolymerization. The Cl 2p core-level spectra of the graft-modified films show a spin-orbit split doublet (Cl 2p_{3/2} and Cl 2p_{1/2}) with peaks at 200 and 201.6 eV

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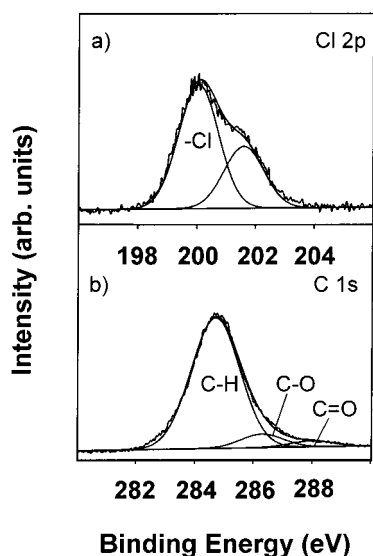


Figure 1. XPS (a) Cl 2p and (b) C 1s core-level spectra of VBC-graft copolymerized LDPE film (sample III in Table 1).

attributable to covalent Cl^{27,28} (Figure 1a). The C 1s core level spectra of the films show a major peak at 284.6 eV consistent with C–H bonds and two smaller peaks at binding energies greater than 286 eV attributed to oxidized species²² (Figure 1b). The VBC graft copolymer concentration on the films (defined as moles of VBC per mole of LDPE repeat unit) is estimated from the XPS analysis according to the equation

$$\frac{M_{\text{VBC}}}{M_{\text{ethylene}}} = \frac{\text{peak area of Cl 2p}}{(\text{peak area of total C 1s} - 9 \times \text{peak area of Cl 2p})/2}$$

where the peak area has been corrected with the appropriate sensitivity factors and the stoichiometric factors of 9 and 2 are introduced to account for the nine C atoms per VBC unit and two C atoms per repeating units of the LDPE substrate. The graft concentration as calculated above is used as a basis for comparing the results obtained under different conditions.

(i) Solution Grafting Technique. The effect of monomer concentration on the grafting efficiency of the solution grafting technique can be seen in Table 1a. An increase in VBC monomer concentration from 1 to 5 vol % results in a corresponding increase in the graft concentration of the films from 0.03 to 0.09. However, the graft concentration decreases with further increase in monomer concentration beyond 5 vol %. A high VBC concentration may have been unfavorable for graft copolymerization due to a greater extent of homopolymerization of the VBC in the solution. This will in turn inhibit the propagation of the surface-grafted polymers by acting as a competitive reaction and/or by retarding the diffusion of monomers to the film surface. The VBC graft copolymer concentration also depends on the solvent used for grafting. On the basis of the results obtained using five different solvents (Table 1b), the graft copolymerization of VBC on LDPE films appears to be more effectively carried out in protic solvents such as 2-propanol and ethanol. The lower graft concentration obtained when aprotic solvents (THF,

Table 1. VBC Graft Copolymer Concentration on 90 s Argon Plasma Pretreated LDPE Films Subjected to Solution Grafting Using (a) Different Monomer Concentrations and (b) Different Solvents

(a) Solvent: Ethanol		
sample	monomer concn (vol %)	VBC graft concn
I	1	0.03
II	2	0.04
III	5	0.09
IV	10	0.05
(b) Monomer Concentration: 2 vol %		
sample	solvent	VBC graft concn
V	ethanol	0.04
VI	2-propanol	0.08
VII	THF	0.01
VIII	xylene	0.03
IX	50–50 vol % DMF/ethanol	0.03

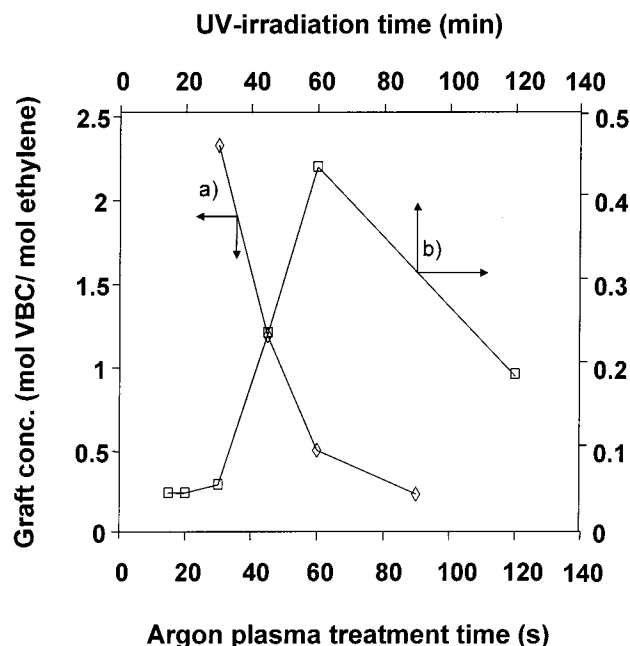


Figure 2. VBC graft copolymer concentration of films subjected to (a) different periods of argon plasma treatment time and 45 min of UV-induced graft copolymerization time and (b) 90 s of argon plasma treatment time and different periods of UV-induced graft copolymerization time.

xylene, and DMF) are used may be due to the higher solubility of the monomer in these solvents, which results in the stabilization of the monomer and thus the lowering of its reactivity.

(ii) Quartz Plate Grafting Technique. The LDPE films used in the quartz plate grafting technique are subjected to different periods of argon plasma pretreatment time. The effect of argon plasma pretreatment time on the graft concentration of the films is shown in Figure 2a. Shorter exposure of the LDPE films to argon plasma is observed to give higher graft concentrations. Studies carried out by Suzuki et al. suggest that longer exposure of the LDPE to argon plasma does not in fact result in larger amounts of peroxide formed on the substrate surface.²⁵ It was postulated that intensive cross-linking of the polymer radicals upon prolonged plasma treatment results in a lower amount of radicals available for peroxide formation when the substrate is subsequently exposed to air. The smaller amount of peroxide on the film surface results in a lower degree of graft copolymerization and thus a lower graft concentration of VBC. At a plasma

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pretreatment time of 90 s, a variation of the VBC-graft copolymerization time from 15 to 120 min results in a maximum graft concentration of 0.44, at a corresponding graft copolymerization time of 60 min (Figure 2b). The initial increase in graft concentration with increase in UV-irradiation time is as expected. The sudden drop in graft concentration of the film subjected to 120 min of irradiation-induced graft copolymerization might be a result of the physical detachment of surface graft copolymerized polymers during the removal of the quartz plates. The strong adhesion of the VBC-graft copolymerized film to the quartz plates after 120 min of graft copolymerization does not diminish even after prolonged immersion in DMF. As such, the plates had to be forced apart by physical means and some graft-copolymerized polymers may have been cleaved in the process.

A comparison of the values in Table 1 and Figure 2 shows that the quartz plate grafting technique gives, on average, a much higher graft concentration of VBC as better contact between the monomer and the polymeric substrate is afforded by this technique. The solution grafting technique, however, has the advantage of producing more homogeneously graft copolymerized films. The extraction of the densely VBC-graft copolymerized films from the quartz plates in the quartz plate grafting technique often results in visually obvious patches of uneven grafting.

Formation of Viologen Moieties. The reaction of the VBC-graft copolymerized films with 4,4'-bipyridine and benzyl chloride or dichloro-*p*-xylene results in the formation of viologens. This is observed as the emergence of a N 1s signal in the XPS analysis of the films. The N 1s core level spectra of the viologen-grafted films can be fitted with three peaks. The peak at 401.7 eV is assigned to the positively charged nitrogen, that at 399.5 eV is attributed to the viologen radical cation formed during X-ray excitation in the analysis chamber,¹⁰ and finally the peak at 398.6 eV is assigned to the unreacted imine nitrogen of the pyridine rings.²⁹ The Cl 2p core-level spectra of the viologen-grafted films are deconvoluted into two spin-orbit split doublets with binding energies for the Cl 2p_{3/2} peaks at 197 and 200 eV attributable to ionic and covalent Cl(-Cl) respectively.^{27,28} An indication of the approximate amount of viologen grafted on the LDPE surface can be inferred from the [N]/[C] ratio and the ratio of the covalent Cl to the total Cl ([-Cl]/[Cl_T]) of the films. A high [N]/[C] ratio would imply the successful introduction of a large number of the bipyridine units while a high [-Cl]/[Cl_T] ratio would indicate that a number of VBC graft copolymer units did not react with the bipyridine units.

(i) Method 1. The reaction of sample III with 4,4'-bipyridine (step 1) at 70 °C is monitored as a function of time using UV-visible absorption spectroscopy. An increase in the absorbance at approximately 260 nm is seen with reaction time, consistent with the formation of benzyl viologen.²⁶ The similarity in absorbances of the spectra obtained after 15 and 20 h indicates that a reaction time of 20 h is sufficient for this reaction step. UV-visible absorption spectroscopy is also used to monitor the subsequent reaction of the film with benzyl chloride (step 2) at 70 °C. In a similar manner, it can be deduced that 10 h is sufficient for the completion of the step 2 reaction. Figure 3a shows the N 1s core level spectrum of sample III after step 1 (20 h of reaction). The [-N=]/[N] ratio of this film is substantially less than 0.5, indicating that most bipyridine rings are already diquaternized after this

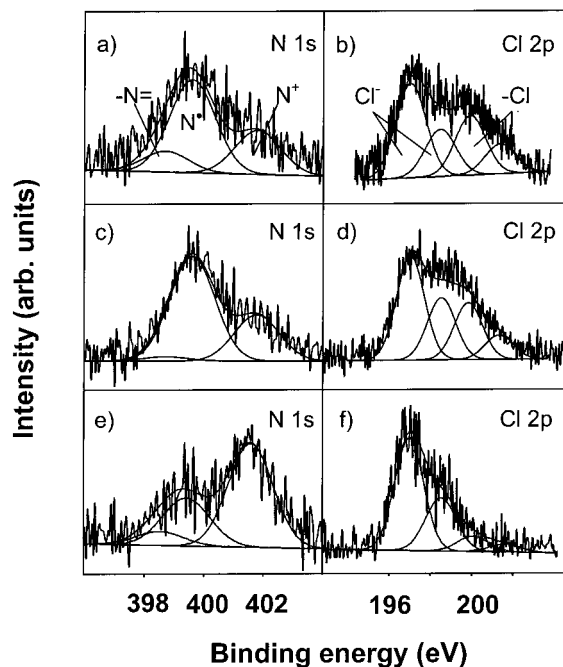


Figure 3. XPS N 1s and Cl 2p core-level spectra of sample III after 20 h of reaction with 1 wt % 4,4'-bipyridine at 70 °C (a and b), sample III after 20 h of reaction with 1 wt % 4,4'-bipyridine at 70 °C, followed by another 20 h of reaction with 5 vol % benzyl chloride at 70 °C (c and d), and sample III reacted with an equimolar mixture of 4,4'-bipyridine and dichloro-*p*-xylene (0.024 M of each reactant) at 60 °C for 20 h (e and f).

Table 2. [N]/[C] Ratio of Sample III Reacted with 4,4'-Bipyridine and Benzyl Chloride (20 h reaction time for both steps 1 and 2) at Different Temperatures

step 1 temp (°C)	step 2 temp (°C)	[N]/[C]
40	40	0.03
70	70	0.05
90	90	0.05

first reaction step. The Cl 2p core level spectra of the film (Figure 3b) shows a substantial amount of covalent chloride present even though the bipyridine was added in excess during reaction. The [-Cl]/[Cl_T] ratio of the film is calculated to be 0.40; hence, approximately half the amount of graft copolymerized benzyl chloride groups have reacted to form viologens on the film surface. This incomplete reaction of the benzyl chloride groups in VBC may be due to steric hindrance from the benzene rings. Figure 3c shows the N 1s core level spectrum of sample III after both steps 1 and 2 (20 h of reaction for each step). The decrease in [-N=]/[N] from 0.13 after step 1 to 0.02 after step 2 shows the role of this step in completely diquaternizing the bipyridine rings. The diquaternizing of the rings by the benzyl chloride added in step 2 also introduces more ionic chloride to the film and [-Cl]/[Cl_T] of the film decreases to 0.32.

To monitor the effect of temperature on steps 1 and 2, the reactions of sample VI with 4,4'-bipyridine and benzyl chloride were carried out at 40, 70, and 90 °C. The [N]/[C] ratios of these films are indicated in Table 2. While the conversion of the graft-copolymerized benzyl chloride groups to viologens is less when the reactions are carried out at 40 °C, the [N]/[C] ratio of the films reacted at 70 and 90 °C does not differ much.

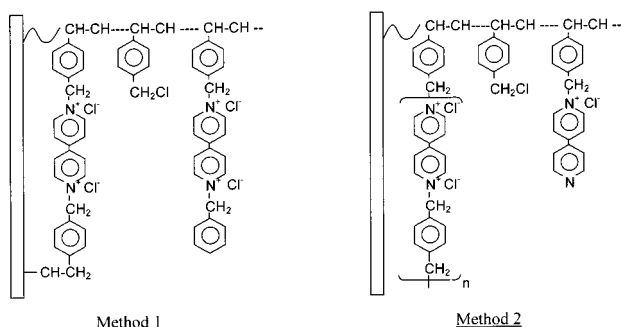
(ii) Method 2. Method 2 is carried out using the bifunctional dichloro-*p*-xylene instead of the monofunctional benzyl chloride. It is envisioned that this reaction would cause chain lengthening of the pendent groups to

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Table 3. [N]/[C] Ratio of Viologen-Grafted Films Synthesized Using Different Reaction Conditions

film	VBC graft concn	method of viologen synthesis ^a	[N]/[C]
A	0.06	1	0.05
B	0.09	1	0.05
C	0.09	2	0.08
D	0.20	1	0.06
E	0.22	2	0.12
F	0.51	1	0.09

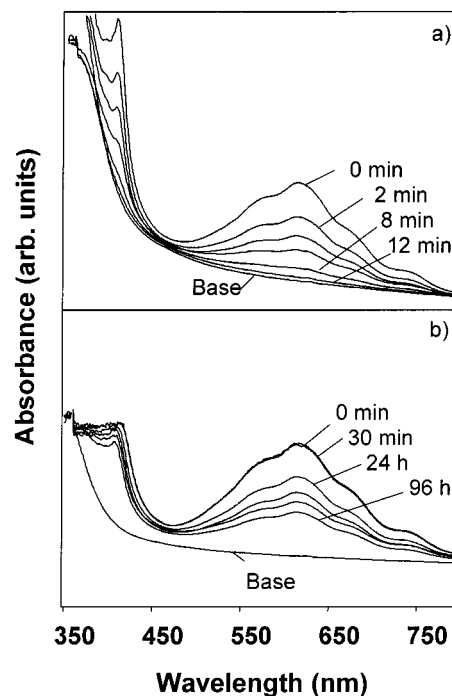
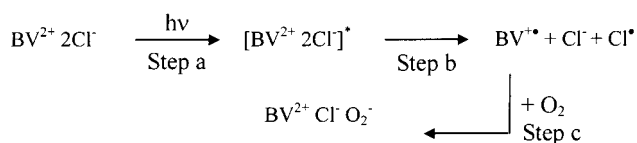
^a Films reacted using method 1 are subjected to 20 h of reaction at 70 °C for both steps 1 and 2. Films reacted using method 2 are subjected to 20 h of reaction at 60 °C.

Chart 1

result in longer, ionene-type polyviologens. Figure 3e shows the N 1s core level spectrum of sample III after reaction in an equimolar mixture of 4,4'-bipyridine and dichloro-*p*-xylene for 20 h. The presence of the peak at 398.6 eV suggests that some bipyridine rings serve as end groups in this sample, resulting in a small amount of unreacted imine nitrogens. Compared to the film from method 1, the XPS Cl 2p core-level spectrum of the film from method 2 shows an increase in the proportion of Cl⁻ anions (comparing Figure 3f and Figure 3d). This may be indicative of the lengthening of the pendent groups to result in a greater amount of viologen moieties and, hence, a corresponding amount of counteranions. The $[-Cl]/[Cl_T]$ ratio of this film is 0.18 compared to 0.32 for the film prepared by method 1.

Table 3 gives the [N]/[C] ratio of some of the viologen-grafted films prepared by the two different methods. Although there are many possible and complicated structures of the viologen polymers on each film, some deductions regarding the effects of VBC graft concentration and the different methods (methods 1 and 2) on the subsequent viologen graft concentration can be made by comparing the [N]/[C] ratio of films. In general, for the same method of viologen synthesis, an increase in VBC graft concentration results in a greater amount of viologen grafted and a higher [N]/[C] ratio. For films having similar VBC graft densities, a higher [N]/[C] ratio and lower $[-Cl]/[Cl_T]$ ratio is obtained when the film is reacted using method 2 as compared to method 1, which suggests that the latter is effective in elongating the pendent groups on the grafted viologen polymers. Hence, from the XPS analysis, possible structures of the surface-grafted viologens (not in stoichiometric proportions) using methods 1 and 2 are deduced (Chart 1).

The AFM images (surface view) of pristine, argon plasma treated, VBC-graft copolymerized, and viologen-grafted LDPE films were studied. The surface of the pristine LDPE film is relatively smooth, with a root-mean-square (rms) roughness of 6.3 nm. The rms roughness of the film increases to 9.7 nm after plasma treatment, 9.9 nm after VBC graft copolymerization, and further in-

**Figure 4.** UV-visible absorption spectra of film D irradiated and bleached in (a) air and (b) vacuum.**Scheme 3**

creases to 14.0 nm with the formation of the viologen polymers on the film surface according to method 2. Comparing the AFM images of the pristine and viologen-grafted LDPE films, viologen polymers can be seen to have amassed along the vertical direction perpendicular to the film's surface.

Film Properties. The response of the viologen-grafted films to photoirradiation and their subsequent bleaching in air as well as in a vacuum were monitored using UV-visible absorption spectroscopy. A typical bleaching profile of the viologen-grafted films in air is shown in Figure 4a. The "base" spectrum is for the film before irradiation, and no absorption bands are seen in the range of 350 to 800 nm. The "0 min" spectrum denotes the spectrum of the film after 3 min of irradiation using the 1-kW Hg lamp. The emergence of the absorption bands at 615 and 410 nm indicates the formation of viologen radical cations^{30–32} (denoted as BV⁺). The formation of the viologen radical cation upon photoirradiation is well documented and is caused by the transfer of an electron from the counteranion to the viologen dication (BV²⁺) as depicted in steps a and b of Scheme 3.^{32–34} After the 3-min irradiation period, the film is allowed to bleach in the dark. In the presence of air, the reaction of the viologen radical cations with O₂

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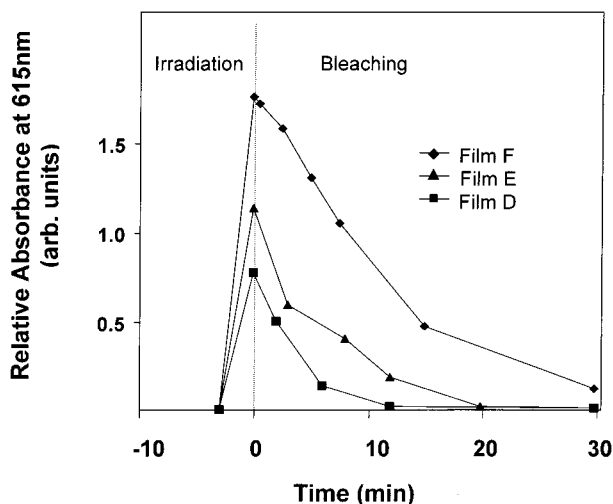


Figure 5. Response of viologen-grafted LDPE films with different $[N]/[C]$ ratios to irradiation and the subsequent bleaching in air.

regenerates the viologen dication (step c) and this reaction has been known to be particularly fast.³⁵ This results in the decrease in the intensity of the absorption bands at 615 and 410 nm. The spectrum of the film eventually approaches that of the base when all radical cations are converted back to the dications. These changes in the viologen-grafted films upon irradiation can also be followed visually as the films change from pale yellow to blue upon irradiation, and the blue coloration fades as the films bleach.

The amount of viologen grafted on the films affects the response to irradiation, as well as the bleaching, as shown in Figure 5. The relative absorbance of the film is calculated as the absolute absorbance at any one time minus the absolute absorbance of the base. On comparison of the response of films D ($[N]/[C] = 0.06$) and F ($[N]/[C] = 0.09$), an increase in the amount of viologen grafted results in a corresponding increase in the amount of viologen radical cation formed (seen as an increase in relative absorbance at 0 min) and, hence, the time required for complete subsequent bleaching. The higher absorbance of film E (similar VBC graft concentration as film D, but $[N]/[C] = 0.12$) compared to film D again confirms the efficiency of method 2 in increasing the amounts of viologens grafted onto a VBC-graft copolymerized film as compared to method 1. The role of oxygen in the bleaching of the films can be seen in Figure 4b. When film D is irradiated in a vacuum and subsequently left in the dark, the viologen radical cations persist even after 4 days, whereas the radical cations in a similar piece of film irradiated and bleached in air have completely converted back to the dications after 12 min (Figure 4a). The stability of the viologen radical cations in the absence of air is expected since the fast reaction of the radicals with O_2 is eliminated. The slow decrease in absorbance of the 615 nm peak with time in Figure 4b may be due to the back electron transfer from the viologen radical cation to the halide radical³⁴ and also to any O_2 from the slow diffusion of air into the vacuum cell.

The response of the viologen-grafted films to irradiation also depends on the intensity of the irradiation. Figure 6a shows the response of a viologen-grafted film to the full intensity of the 1-kW Hg lamp, as well as those when exposed to 20% and 50% transmission of the irradiation, respectively. The peak absorbances obtained at 100% and

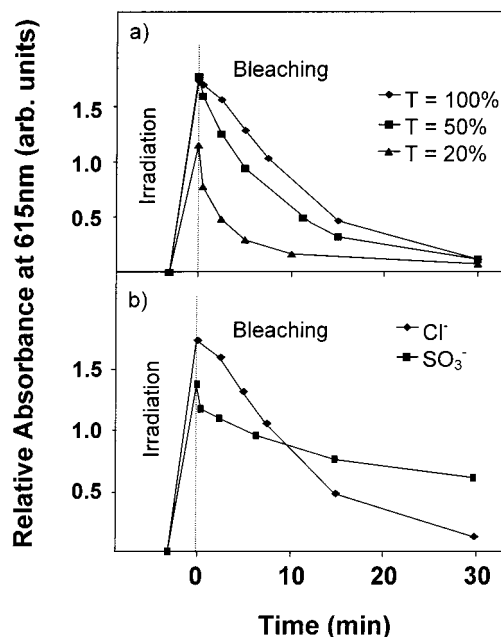


Figure 6. Response and bleaching of (a) viologen-grafted films exposed to different intensities of irradiation in air where T is the transmittance of the optical density filter and $T = 100\%$ implies that the irradiation is not attenuated and (b) viologen-grafted films with different counteranions.

50% transmission are rather similar, indicating that the amounts of viologen cation radicals formed do not vary much. At 20% transmission, the amount of viologen radicals formed on the film is almost half of that formed upon full irradiation, assuming that the absorbance at 615 nm is directly proportional to the concentration of viologen cation radicals. The rate of bleaching of the less intensely irradiated film is also faster, as seen by the steeper decrease in relative absorbance of the film when placed in the dark.

A fourth factor that influences the response and bleaching of the viologen films is the counteranion on the viologen moieties. XPS analysis of film F subjected to ion exchange with poly(vinylsulfonic acid, sodium salt), NaSS, for 3 h shows a complete disappearance of the Cl 2p signal and the emergence of a S 2p signal at 167.4 eV (attributable to the SO_3^- anion²⁷), indicating that anion exchange has been successfully achieved. A comparison of the response and bleaching of the original viologen-grafted film (Cl^- as anion) and the NaSS-exchanged film is shown in Figure 6b. While the sensitivity of the original viologen-grafted film (with Cl^- counterions) to irradiation is greater (a greater amount of viologen radical cations is formed), the poly(vinylsulfonate) anions impart a greater stability to the radical cations formed during irradiation through electron delocalization³⁵ and this results in a longer bleaching time.

To examine the write-erase efficiency of the viologen-grafted films, film F is subjected to cycles of alternate irradiation (100% transmission) and bleaching in air. Figure 7a shows the cycling results of the sample for eight cycles. The amount of viologen radicals formed decreases after each cycle, as indicated by the decrease in absorbance at 615 nm. Figure 7b shows the UV-visible absorption spectra of the sample before irradiation and after eight cycles of irradiation and bleaching. The absorption spectrum of the cycled film has shifted upward in the 400–500 nm region.³⁶ The film is also visually observed

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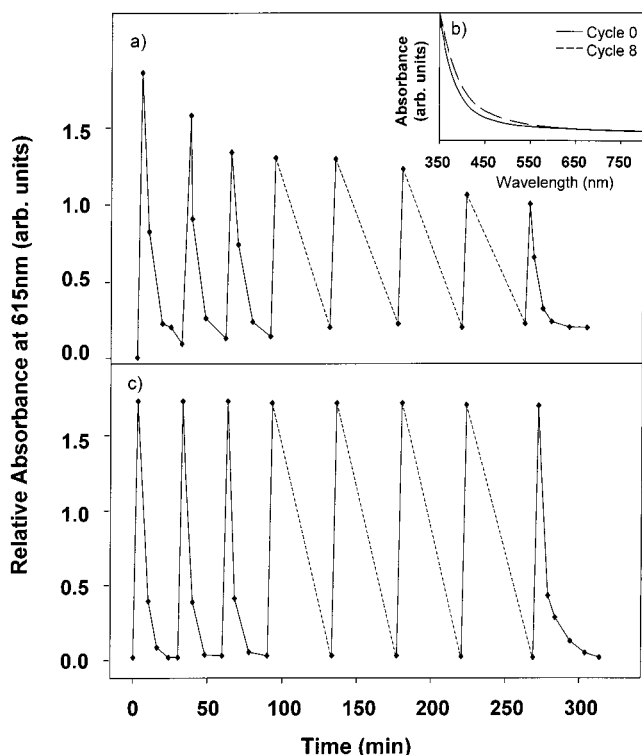


Figure 7. (a) Response of film F subjected to eight cycles of alternate irradiation ($T = 100\%$) and bleaching in air, (b) UV-visible absorption spectra of film F before irradiation and after eight cycles of irradiation and bleaching, and (c) response of film F subjected to eight cycles of alternate irradiation ($T = 50\%$) and bleaching in air.

to turn slightly orange. Similar observations have been reported in electrochromic display devices containing viologens after prolonged cycling,^{37–39} and the yellowish-brown product has been attributed to the dimerized viologens. The observed color change may also be due to the formation of the viologen radical cation dimer, which can occur to an appreciable extent in the solid state.³⁷ With each irradiation cycle, this conversion of a small amount of the viologen radical cations to the other viologen species occurs and this results in the falling absorbance of the film at 615 nm (the film absorbance decreases by 56% after eight cycles). For a similar piece of film cycled using 50% transmission from the 1-kW Hg lamp, a better

write-erase efficiency is observed (Figure 7c). This may be due to the lesser amount of the viologen radical cations being converted to the other species in each cycle with the reduction in the amount of energy supplied to the film. The film absorbance at 615 nm decreases by less than 2% after eight cycles of irradiation and bleaching.

Viologen Grafting on Other Polymeric Substrates.

Besides using LDPE as the polymeric substrate, viologen moieties were also successfully grafted on polypropylene and nylon using the surface graft copolymerization technique presented above. For similar grafting conditions as film D (60 s plasma treatment, 20 min UV-induced graft copolymerization using the quartz plate technique, and reactions to form viologen moieties employing method 1, at 70 °C and 20 h for both reaction steps) an $[N]/[C]$ ratio of 0.05 and 0.07 is obtained for viologen-grafted polypropylene and nylon, respectively. These values are very similar to that obtained with the LDPE film ($[N]/[C] = 0.06$), and these viologen-grafted polypropylene and nylon films also respond in a similar manner upon irradiation, to result in the formation of the blue viologen radical cation.

Conclusion

Viologen moieties have been successfully grafted on LDPE, nylon, and polypropylene via a surface graft copolymerization technique. This technique involves, first, the graft copolymerization of VBC on the polymeric substrate, followed by reactions to form viologens on the VBC-graft copolymerized film. The amount of viologen grafted depends to a large extent on the graft concentration of VBC, which in turn depends on the plasma pretreatment time of the film, the UV-induced graft copolymerization time, and, in the case of solution grafting, the solvent and monomer concentration used. The introduction of a bifunctional species such as dichloro-*p*-xylene in the viologen synthesis reaction can also increase the amount of viologen grafted by allowing chain elongation on the viologen polymer. Photoirradiation of the pale yellow viologen-grafted films results in the formation of the intensely blue viologen radical cation. Bleaching of the irradiated films occurs when placed in the dark. The response and bleaching behavior of the viologen-grafted films depends on the amount of viologen grafted, the light intensity, the counteranion associated with the viologen moieties, and the presence of oxygen. Prolonged cycling of the films is postulated to result in the formation of a small amount of other viologen derivatives.

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